

Python Interview Preparation

Written by: Muhammad Nouman

1. What is python ?

Answer : Python ek high-level, interpreted, object-oriented programming language hai. Ye simple syntax ki wajah se famous hai aur web development, AI, automation, scripting aur data science me use hoti hai.

Features:

- Easy syntax
- Readable code
- Large community support
- Cross-platform
- Huge libraries support

Example:

```
print("Hello World")
```

2. Python Interpreted Language Kyu Kehlati Hai?

Answer: Python interpreted language hai kyun ke iska code line by line execute hota hai compiler ki tarah pehle pure program ko machine code me convert nahi karta.

3. What are Data Types in Python?

Answer: Python me different types ka data store karne ke liye built-in data types hote hain. Inhein different categories me divide kiya jata hai.

Types of Data Types in Python

Category	Data Types	Example
Numeric	int, float, complex	10, 4.12, 2+3j
Sequence	list, tuple, range	[1,2], (1,2)
Text	String	"Hello"
Set	Set	{1,2,3}
Mapping	Dict	{"a" : 1}
Boolean	bool	True, False
Binary	bytes, bytearray, memoryview	b"hello"

Python may built-in data types wo hote hain jo different types ka data store krte hain.

4. Variables kia hote hain?

Answer: variable memory location ka nam hote hain jis may data store hota hai

5. What is mutable and immutable in python?

Answer:

i. mutable wo data types hote hain jin ki value ko change kia ja skta hai

Example:

- List
- Dict
- Set

ii. immutable data types wo hote hain jin ki value ko change nahi kia ja skta

Example:

- Typle
- String
- Int

6. What is the difference between list and tuple?

Answer:

i. List mutable hai isko square brackets may define kia jata hai or ye tuple se thori slow hoti hain

ii. tuple immutable hai or isko parentheses may define kia jata hai tuple list se thori fast hoti hai

7. What is type casting in python?

Answer: aik data type ko dusre data type may convert krna type casting kehlata hai python may do tarha ki type casting hoti hai

i. Implicit

Python automatically convert krta hai jb zarorat ho

ii. Explicit

Hum manually conversion krte hain

8. Operators in python ?

Answer:

i. Arithmetic operators

+, -, *, /, //, %, **

ii. Comparison Operators

==, !=, >, <, >=, <=

iii. Logical Operators

And, or, not

9. What is a function?

Answer: function reusable block of code hota hai jo specific task ko perform krta hai

Example:

```
def greet(name):  
    return f"hello{name}"
```

10. What is difference between parameter and arguments:

Answer: **Parameters** function k andar define hote hain (Variables)

Arguments function ko call krte hain waqt pass kiy jate hain (values)

Types of Arguments

- Positional Arguments
- Keyword arguments
- Default arguments
- variable-length Arguments

Examples:

i. Positional Arguments

Defination: jaha order tarteeb matter krta ho

```
def student(name, age, city):  
    print(f"{name} is {age} years old from {city}")
```

Sahi - Order match kar raha hai

```
student("Ali", 20, "Karachi")
```

Output: Ali is 20 years old from Karachi

Ghalat - Order mismatch

```
student("Karachi", "Ali", 20)
```

Output: Karachi is Ali years old from 20 (meaningless)

ii. Keyword Arguments

Defination: jaha parameter name explicitly likhte hain

```
def student(name, age, city):  
    print(f"{name} is {age} years old from {city}")
```

Order koi bhi ho sakta hai

```
student(age=20, city="Lahore", name="Sara")
```

Output: Sara is 20 years old from Lahore

Mix bhi kar sakte hain (positional pehle, keyword baad mein)

```
student("Ahmed", city="Islamabad", age=22)
```

Output: Ahmed is 22 years old from Islamabad

iii. Default Arguments

Defination: jaha parameter ki pehle se value set ho

```
def greet(name, message="Hello"):  
    print(f"{message}, {name}!")
```

Bina message diye - default use hoga

```
greet("Ali")
```

Output: Hello, Ali!

Message diya - custom use hoga

```
greet("Sara", "Assalam Alaikum")
```

Output: Assalam Alaikum, Sara!

Multiple default arguments

```
def student(name, age=18, city="Karachi"):
```

```

print(f"{name} is {age} years old from {city}")

student("Ahmed")           # age=18, city=Karachi
student("Ahmed", 20)        # city=Karachi
student("Ahmed", 20, "Lahore") # sab custom

```

Iv. Variable-Length Arguments

*A) argos (Positional Variable-length)

Definition: jahan jaha multiple values bina key k pass krni ho

```

def sum_all(*args):
    print(f"Numbers received: {args}") # args is a TUPLE
    total = 0
    for num in args:
        total += num
    return total

```

```

# 2 arguments
print(sum_all(5, 10))
# Output: 15

```

```

# 5 arguments
print(sum_all(1, 2, 3, 4, 5))
# Output: 15

```

```

# 0 arguments bhi de sakte hain
print(sum_all())
# Output: 0

```

**B) kwargs (keyword variable-length)

Definition: jahan multiple key-value pairs pass krni ho

```

def print_details(**kwargs):
    print(f"Data received: {kwargs}") # kwargs is a DICTIONARY
    for key, value in kwargs.items():
        print(f"{key} = {value}")

```

```

# Multiple keyword arguments
print_details(name="Ali", age=20, city="Karachi", profession="Developer")
# Output:
# name = Ali
# age = 20
# city = Karachi
# profession = Developer

```

```

# 0 arguments bhi de sakte hain
print_details()
# Output: (kuch print nahi hoga)

```

11. What is the lambda function?

Answer : lambda aik anonymous single-line function hota hai

Examples:

```
square = lambda x: x **2  
print(square(5)) # output 25
```

12 what is *args and **kwarg?

Answer:

i. *args

multiple Positional arguments store karta hai

ii. **kwargs

Multiple keyword arguments receive karta hai

13. What is scope?

Answer: scope variable ki accessibility define karta hai. Python may main do scope hote hain

i. Local Scope variable jo function k andar define hota hai, siraf us function k andar accessible hota hai

ii. Global Scope variable jo function k bahir define hota hai, puri file may accessible hota hai

Important point: agr function k andar global variable ko modify krna hoto 'global' keyword use hota hai warna python usay local variable assume kr leta hai.

Examples:

Predict the output:

```
x = 100      # Global
```

```
def func1():  
    x = 50    # Local (different from global x)  
    print(x)  # ?
```

```
def func2():  
    global x  
    x = 200   # Modifying global x  
    print(x)  # ?
```

```
func1()  
func2()  
print(x)    # ?
```

Answers: 50, 200, 200

Common interview trap question

```
my_var = 10
```

```
def test():
```

```
print(my_var) # Line 1
my_var = 20   # Line 2
print(my_var) # Line 3
```

test()

Question: output kia hoga

Answer: unboundlocalError

Reason: python function k andar my_var = 20 dekh kr usay local variable treat krta hai line 1 pe print krte waqt local variable exist nahi krta is lia error ata hai.

Solution:

```
my_var = 10
```

```
def test():
    global my_var # Add this line
    print(my_var) # 10
    my_var = 20   # modifies global
    print(my_var) # 20
```

test()

14. What is recursion?

Answer: jb function khud ko call kre to recursion kehlata hai is k do parts hote hain

- Base case: (rukne ki condition)
- Recursive case (khud ko call krne ki condition)

Example:

```
def factorial(n):
    if n == 1:
        return 1
    return n * factorial(n-1)
```

15. What is exception handling

Answer: program crash hone se bachane k liye exceptions handle krte hain

Keywords:

- try
- except
- finally
- raise